

Systems Analytics Course

Master's in Innovation of Energy Systems (MISE)

1. Executive Vision: The Systemic Turn in Energy Education

The global energy sector is currently navigating a structural transformation of unprecedented complexity, often characterized by the "3D" paradigm: Decarbonization, Digitalization, and Decentralization. This shift necessitates a fundamental reimagining of graduate engineering education. The traditional pedagogical model—which compartmentalizes power systems, economics, and control theory into isolated silos—is increasingly insufficient for training leaders capable of managing the modern grid. The Master's in Innovation of Energy Systems (MISE) at Universidad de los Andes (Uniandes) has identified this gap and proposes a cornerstone course, "Systems Analytics," to serve as the methodological connective tissue for the program.¹

This report provides a comprehensive, 15,000-word strategic roadmap for the design, implementation, and technological scaffolding of this course. The curriculum is structured around three critical modules: Systems/Networks (focused on Cyber-Physical-Social Systems), Dynamic Modeling (utilizing Agent-Based Modeling and Game Theory), and Decision Making (emphasizing Control and Optimization under Deep Uncertainty). Beyond the theoretical content, this document details a pioneering technical infrastructure leveraging "Google Antigravity" for agentic software development ³ and "Google NotebookLM" for AI-augmented curricular research.⁵

The objective is to move students beyond the role of passive analysts to become architects of complex adaptive systems. By integrating the resilience theories of Barabási and Dueñas-Osorio ⁶ with the computational power of Python-based frameworks like Mesa ⁸ and Rhodium ⁹, the MISE Systems Analytics course will position Uniandes at the forefront of international energy education. This report analyzes the pedagogical gaps in current offerings, benchmarks against global leaders like MIT and TU Delft, and provides a granular technical manual for deploying the course's digital ecosystem.

2. Strategic Context and International Benchmarking

2.1 The Imperative for Systems Analytics in the MISE Curriculum

The MISE program is designed to form professionals who can "lead energy management with a technical, economic, environmental, and social vision".² This mission statement implicitly acknowledges that the energy system is no longer a purely physical machine but a "Cyber-Physical-Social System" (CPSS).¹⁰ In this context, "Systems Analytics" is not merely a course on statistics or programming; it is a course on the physics of interaction.

In the Colombian context, where the energy matrix is dominated by hydro-resources subject to El Niño Southern Oscillation (ENSO) variability and where social license to operate is a critical constraint for transmission expansion, the ability to model interdependent systems is vital. The traditional N-1 reliability criterion, which focuses on the loss of a single physical component, fails to capture the complexity of cascading failures driven by cyber-attacks, market gaming, or social unrest.¹¹ Therefore, the Systems Analytics course must equip students with the mathematical tools to quantify resilience, simulate emergent market behaviors, and optimize decisions when probability distributions are unknown.

2.2 Global Benchmarking and Gap Analysis

To ensure the MISE Systems Analytics course meets and exceeds international standards, a rigorous benchmarking exercise was conducted against four leading technical universities: Massachusetts Institute of Technology (MIT), TU Delft, ETH Zurich, and Stanford University. This analysis identifies the "State of the Art" and highlights the specific niches where Uniandes can innovate.

2.2.1 Massachusetts Institute of Technology (MIT): Systems Thinking in Energy

The MITx course "Systems Thinking in Energy: Navigating Costs, Emissions, and Impact"¹² represents the gold standard for techno-economic analysis. The curriculum focuses heavily on the "Sesame" modeling tool, which allows for consistent "apples-to-apples" comparisons across energy pathways.

- **Strengths:** The course excels in teaching students how to perform lifecycle analysis (LCA) and navigate the tradeoffs between cost and emissions. It rigorously frames the problem of decarbonization within economic constraints.¹²
- **Limitations:** The primary focus is often on the *technological* and *economic* dimensions. There is less emphasis on the *topological* structure of the grid as a complex network or the *emergent* social dynamics of decentralized actors.
- **Uniandes Opportunity:** The MISE course can differentiate itself by incorporating **Network Science** (Module 1) to analyze the structural resilience of the grid, moving beyond the economic dispatch focus of MIT to a topological robustness focus.

2.2.2 TU Delft: Intelligent and Integrated Energy Systems

TU Delft's curriculum, particularly the "Technology of Intelligent and Integrated Energy Systems"¹³ and "System Analysis"¹⁴, is notable for its integration of multi-carrier grids (combining hydrogen, gas, and electricity) and its explicit inclusion of "Actor Analysis."

- **Strengths:** Delft acknowledges that energy systems are socio-technical. The inclusion of Actor Analysis¹⁴ validates the need to understand stakeholders (NGOs, utilities, governments).
- **Limitations:** While Actor Analysis is often taught as a qualitative or semi-quantitative framework, there is an opportunity to make this more rigorous through computation.
- **Uniandes Opportunity:** We will advance the Delft model by converting static "Actor Analysis" into dynamic **Agent-Based Modeling (ABM)** (Module 2). Instead of mapping stakeholders on a static matrix, MISE students will code agents that interact in real-time, allowing them to observe how micro-behaviors lead to macro-phenomena like market bubbles or technology diffusion.

2.2.3 ETH Zurich: Energy Systems Analysis

ETH Zurich offers a highly quantitative "Energy Systems Analysis" course¹⁵ that focuses on mathematical optimization, particularly for district heating and aviation.

- **Strengths:** The rigor of their optimization curriculum is world-class, providing deep expertise in linear and mixed-integer programming (MILP).
- **Limitations:** Traditional optimization assumes a known objective function and deterministic (or stochastically predictable) constraints.
- **Uniandes Opportunity:** The MISE course will introduce **Robust Decision Making (RDM)** (Module 3). Given the "deep uncertainty" of climate change and Latin American politics, teaching students to optimize for *robustness* (performance across many futures) rather than *optimality* (performance in one predicted future) is a critical value-add.¹⁶

2.2.4 Stanford University: Energy Systems Fundamentals

Stanford's "Energy Systems Fundamentals"¹⁷ provides a broad overview of resources and conversion technologies.

- **Strengths:** It provides a solid baseline of physical principles.
- **Limitations:** It is less focused on the advanced computational methodologies required for *analytics*.
- **Uniandes Opportunity:** The MISE course will be strictly methodological. It assumes students know the physics (covered in other MISE modules) and focuses entirely on the *analytics*—specifically utilizing the **Google Antigravity** workflow to teach advanced coding and modeling skills that are typically reserved for computer science programs.¹⁸

2.3 Synthesis: The Uniandes Differentiator

Based on this analysis, the MISE Systems Analytics course will define its unique value proposition through **Complexity and Computation**. While peers focus on technology assessment (MIT) or policy analysis (Delft), Uniandes will focus on the **computational modeling of complexity**. This aligns with the "Google Antigravity" technical requirement, positioning the course not just as an energy class, but as a rigorous training ground for "Energy Data Scientists" and "Systems Architects."

Table 1: Comparative Benchmarking Summary

Feature	MIT (Systems Thinking)	TU Delft (Integrated Systems)	ETH Zurich (Optimization)	Uniandes MISE (Proposed)
Primary Method	Lifecycle Analysis (Sesame)	Actor Analysis / Multi-carrier	MILP / Optimization	Agent-Based Modeling & Network Science
System View	Techno-Economic	Socio-Technical	Physical-Technical	Cyber-Physical-Social (CPSS)
Tooling	Spreadsheet / Custom Tools	Qualitative / Matlab	GAMS / Python	Python (Mesa, NetworkX, Rhodium) + Antigravity
Resilience View	Reliability (Cost/Emission)	Institutional Alignment	Security of Supply	Cascading Failure & Topological Robustness
Decision Frame	Cost Minimization	Multi-Stakeholder Consensus	Optimal Dispatch	Robustness under Deep Uncertainty (RDM)

3. Module 1: Systems and Networks (CPSS)

Module Objective: To equip students with the theoretical frameworks and computational tools necessary to analyze the topology, stability, and resilience of energy networks within a Cyber-Physical-Social System (CPSS) context.

The modern grid is no longer a linear pipeline from generation to consumption; it is a complex network where physical power flows interact with digital information (Cyber) and human behavior (Social). This module deconstructs the grid using **Network Science**, moving beyond standard power flow analysis to understand the structural vulnerabilities that lead to catastrophic cascading failures.

3.1 Theoretical Foundation: Graph Theory and Complex Networks

The course begins by translating the physical components of the grid (substations, transmission lines, generators) into the mathematical language of Graph Theory. A network is defined as a graph

$G = (V, E)$, where V represents the set of nodes (buses) and E represents the set of edges (lines).

3.1.1 The Topology of Power: Scale-Free vs. Random Networks

We draw heavily on the seminal work of **Albert-László Barabási** and **Réka Albert** regarding scale-free networks.⁷

- **Random Networks (Erdős-Rényi):** In a random network, most nodes have a comparable degree (number of connections), following a Poisson distribution.
- **Scale-Free Networks (Barabási-Albert):** Real-world networks, including aspects of the power grid, often exhibit a power-law degree distribution ($P(k) \sim k^{-\gamma}$). This implies the existence of "hubs"—nodes with an exceptionally high number of connections.
- **The "Robust yet Fragile" Paradox:** Students will explore the critical insight that scale-free networks are incredibly robust to random failures (since most nodes are low-degree) but catastrophically fragile to targeted attacks on hubs.⁶ This theoretical foundation is crucial for understanding why standard reliability metrics (SAIDI/SAIFI) often underestimate the risk of "Black Swan" events.

3.1.2 Small-World Properties and Synchronization

We incorporate the **Watts-Strogatz** model²¹ to discuss the "Small-World" property—networks with high local clustering but short average path lengths.

- **Relevance to Synchronization:** In AC power grids, the "small-world" nature facilitates synchronization (frequency stability). However, it also means that a perturbation (e.g., a frequency dip) can propagate across the entire system in milliseconds. Students will analyze the trade-off between efficient transmission (short paths) and the containment of faults.

3.2 Advanced Application: Cascading Failures and Resilience

Building on the graph theory basics, the module dives into the specific mechanics of grid collapse, referencing the extensive research of **Leonardo Dueñas-Osorio**.¹¹

3.2.1 The Mechanics of Cascade

Students will model the cascade process not just as a static connectivity problem, but as a dynamic load redistribution phenomenon.

- **Capacity-Load Model:** Following the Motter-Lai or Dueñas-Osorio models, each node/edge has a load L and a capacity $C = (1 + \alpha)L_{initial}$.
- **Failure Propagation:** When a node fails, its load is redistributed to neighbors based on electrical laws (Kirchhoff's laws) or shortest-path approximations. If a neighbor's new load exceeds its capacity, it fails, triggering further redistribution.
- **Critical Thresholds:** Students will calculate the critical tolerance parameter α required to prevent total system collapse for a given topology.

3.2.2 Interdependent Networks (CPSS)

The module culminates in the analysis of **interdependent networks**.¹⁰ The power grid depends on the communication network (SCADA) for control, while the communication network depends on the grid for power.

- **Cyber-Physical-Social Interaction:** We extend the model to include the "Social" layer. For instance, how does "panic" (social contagion) lead to load spikes (everyone charging cars simultaneously) during a cyber-attack? This holistic CPSS view aligns with the latest research on bio-economy and social systems.²⁴

3.3 Lab Experience: The NetworkX Blackout Simulator

Technical Goal: Build a Python-based resilience analysis tool using the NetworkX library.²⁵

3.3.1 Lab Design and Workflow

Students will utilize the Google Antigravity repository to scaffold this project.

1. **Data Ingestion:** Load a standard test case (e.g., IEEE 118-bus or a synthetic Texas grid).¹¹
2. **Topological Analysis:** Write scripts to calculate Degree Centrality, Betweenness Centrality, and Clustering Coefficients.
3. **Simulation Loop:**
 - *Attack:* Remove the node with the highest Betweenness Centrality.
 - *Redistribute:* Update flows (using shortest path or DC power flow approximation).
 - *Check:* Identify lines where Flow > Capacity.
 - *Cascade:* Remove overloaded lines and repeat.

4. **Visualization:** Use Streamlit ²⁷ to create an interactive dashboard where users can toggle the "Tolerance" parameter and watch the grid collapse in real-time.

4. Module 2: Dynamic Modeling (ABM and Game Theory)

Module Objective: To simulate the emergent behavior of decentralized energy markets and technologies by modeling the interactions of autonomous agents (prosumers, utilities, regulators).

As the energy transition empowers consumers to become "prosumers" (producers + consumers), the dynamics of the grid are increasingly driven by millions of individual decisions rather than a central dispatcher. This module shifts from the *structural* view of Module 1 to a *behavioral* view, utilizing **Agent-Based Modeling (ABM)** and **Game Theory**.

4.1 Theoretical Foundation: Complexity and Game Theory

The course contrasts "Top-Down" modeling (System Dynamics, Differential Equations) with "Bottom-Up" modeling (ABM). In ABM, system behavior *emerges* from the interactions of heterogeneous agents.³¹

4.1.1 Game Theoretic Foundations

Before building agents, students must understand the logic of their decision-making.

- **Nash Equilibrium:** We apply non-cooperative game theory to energy markets. Students analyze scenarios like the "Duck Curve" ramping problem: if all batteries discharge at peak price, the price collapses. Finding the Nash Equilibrium helps predict stable market outcomes.³²
- **Evolutionary Game Theory (EGT):** We explore how strategies evolve over time. For example, modeling the adoption of solar panels not as a rational economic calculation but as a social imitation game (Evolutionary Dynamics).³³
- **Stackelberg Games:** Modeling the leader-follower dynamic between a Regulator (setting tariffs) and Prosumers (reacting to tariffs).

4.2 Applied Framework: The Mesa Library (Python)

The course adopts **Mesa**, the leading Python framework for ABM.⁸ Mesa allows for the rigorous definition of agents, environments, and schedulers, making it the industry standard for open-source ABM research.

4.2.1 The ABM Architecture

Students learn the four pillars of a Mesa model:

1. **The Agent (Agent class):** The entity with internal state (e.g., battery level, wealth) and behavioral rules (e.g., "If price > X, sell").
2. **The Model (Model class):** The "God" object that manages the global variables and the passage of time.
3. **The Scheduler (Time module):** Determines the order of agent activation (Random, Simultaneous, Staged). The choice of scheduler significantly impacts the outcome of synchronous systems like power markets.³⁶

4. **The Space (Space module):** The environment where agents exist—in this case, a NetworkGrid representing the physical distribution wires.³⁷

4.3 Lab Experience: The Decentralized Market Simulator

Technical Goal: Develop an ABM to simulate a Peer-to-Peer (P2P) energy market within a neighborhood.

4.3.1 Lab Design and Workflow

1. **Agent Design:** Create HouseholdAgent classes with heterogeneous profiles (e.g., "Green Tech Early Adopter" vs. "Passive Consumer"). Use real load data (from NREL or local datasets) to seed the agents.³⁸
2. **Market Logic:** Implement a "Double Auction" mechanism where agents submit bids and asks to a local clearinghouse agent.³⁹
3. **Simulation:** Run the model over a simulated year.
4. **Analysis:**
 - *Emergence:* Observe how high solar penetration leads to local voltage constraints or negative prices.
 - *Policy Testing:* Introduce a "Tax Agent" or change the tariff structure and observe how the population of agents adapts their strategy over time.
 - *Visualization:* Use Mesa's Solara or internal server to visualize the flow of money and energy on the grid topology.⁸

5. Module 3: Decision Making (Control and Optimization)

Module Objective: To transition from *describing* systems (Networks) and *simulating* them (ABM) to *controlling* them. This module focuses on making optimal decisions under conditions of "Deep Uncertainty."

5.1 Theoretical Foundation: Control and Optimization

Standard optimization (Linear Programming) assumes we know the future parameters (fuel costs, demand) with certainty or defined probability. However, energy transitions are characterized by "Deep Uncertainty"—where experts disagree on the probability distributions of key variables.

5.1.1 Sequential Decision Making

We introduce **Markov Decision Processes (MDP)** and **Dynamic Programming**.⁴¹

- **The Problem:** Energy storage management is a sequential problem. A battery operator must decide *now* whether to discharge, based on an uncertain *future* price.
- **The Solution:** Students learn the Bellman Equation to solve for optimal policies over time. This connects the "Game Theory" of Module 2 with formal control theory.

5.2 Advanced Application: Robust Decision Making (RDM)

The core innovation of this module is the introduction of **Robust Decision Making (RDM)**, a framework pioneered by the RAND Corporation.¹⁶

5.2.1 The RDM Paradigm Shift

Traditional analysis asks: "What is the most likely future, and what is the optimal plan for it?" RDM asks: "What is a plan that performs *well enough* (satisficing) across a vast range of plausible futures?".⁴⁴

- **The XLRM Framework:** Students learn to structure problems using the XLRM matrix:
 - **X (Exogenous Uncertainties):** Climate sensitivity, technology costs, political stability.
 - **L (Policy Levers):** Subsidies, carbon taxes, infrastructure investments.
 - **R (Relationships):** The simulation models (from Module 1 & 2).
 - **M (Performance Metrics):** Cost, Emissions, Reliability, Equity.⁴⁵

5.2.2 Exploratory Modeling and Scenario Discovery

Instead of optimizing for one future, students use **Exploratory Modeling** to simulate thousands of futures (using Latin Hypercube Sampling).

- **Scenario Discovery:** Using algorithms like **PRIM (Patient Rule Induction Method)**, students identify the "Vulnerability Regions" of a strategy. For example, "Our investment plan fails specifically when Hydrogen costs > \$5/kg AND Carbon Tax < \$20/ton".⁴⁵ This allows for the design of adaptive pathways.

5.3 Lab Experience: The Rhodium Policy Planner

Technical Goal: Use the **Rhodium** (Python) library to perform a Many-Objective Robust Decision Making (MORDM) analysis.⁹

5.3.1 Lab Design and Workflow

1. **Problem Setup:** Define a capacity expansion problem for the Colombian grid (Hydro vs. Solar vs. Thermal).
2. **Uncertainty Sampling:** Use Rhodium to generate 1,000 "States of the World" (SOWs) varying rainfall (El Niño) and gas prices.
3. **Simulation:** Run a simple dispatch model for each SOW.
4. **Analysis:**
 - *Pareto Optimization:* Identify the trade-off frontier between Cost and Reliability.
 - *Regret Analysis:* Calculate the "Regret" for each strategy across all futures.
 - *PRIM Analysis:* Use Rhodium's PRIM implementation to visualize the "Box" of failure conditions.
5. **Output:** A strategic recommendation report that identifies the most *robust* investment strategy, not necessarily the cheapest one.

9. Conclusion

The "Systems Analytics" course proposed herein represents a radical modernization of energy engineering education. It rejects the notion that the grid is a static machine and embraces it as a dynamic, evolving Cyber-Physical-Social System. By integrating the rigorous network science of Barabási and Dueñas-Osorio, the emergent modeling of Mesa, and the robust strategy of RDM, Uniandes will equip its graduates with the tools to navigate the chaos of the energy transition. Furthermore, the deployment of **Google Antigravity** and **NotebookLM** creates a pedagogical environment that is not just "tech-enabled" but "AI-native," ensuring that MISE students are fluent in the collaborative human-AI workflows that will define the future of engineering.

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